

A Generative Statistical Model of Intent-Conditioned, Multi-Taste Music Preference

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Abstract

We formulate personal musical taste as a probabilistic program $T_u = (\mathcal{O}, \mathcal{O}_u, G_u, P_u)$. Shared parameters $\mathcal{O}$ specify song, language, response, and population distributions; listener parameters $\mathcal{O}_u$ specify personal taste distributions; a dynamic factor graph G_u represents model-relative song representations, multimodal song-region response profiles, listener perceptions, multiple evolving taste states, intent, exposure, and feedback; and a stochastic program P_u performs posterior inference and samples an intent-conditioned ideal-song distribution. Shared response profiles learn several perceptual and psychological descriptions for every song, section, and passage, while listener and intent variables preserve disagreement and change. We prove latent non-identifiability from commercial recordings, sufficient conditions for existence of the target distribution, stochastic-program semantics, a conjugate prompt update, persistence of personalization under a shared prompt, and additive explanation efficiency. A synthetic calculation demonstrates how the same prompt can favor Rihanna’s *Diamonds* for one listener and Anyma’s *After Love* for another. The numerical profiles illustrate inference and are not empirical song measurements.

1 Problem statement

A listener may value a specific passage, vocal technique, production relationship, lyrical interpretation, setting, or momentary intent. Different listeners may assign different perceptual or psychological descriptions to the same passage, and one listener response may change with intent. The system must infer a distribution over desired music for a current intent and retrieve recordings that approximate it. Prompt retrieval, multimodal representation, conversational feedback, and long-term personalization appear in recent work [1, 2, 3]. Our formulation explicitly separates commercial recordings, learned song representations, shared multimodal response profiles, listener-conditioned perception, preference dynamics, and target distributions.

2 Ontology and notation

Symbol	Meaning and status
W_s, L_s, M_s	Observed mastered waveform, lyrics, and catalog metadata.
H_s	Latent multiscale segmentation; $\mathcal{R}(H_s)$ is its finite set of global, section, and passage regions.
X_s	= Model-relative latent song representation with global and region-level states.
$(X_s^{\text{global}}, \{X_{s,r}\}_{r \in \mathcal{R}(H_s)})$	
$W_{s,r}, L_{s,r}$	Waveform and aligned lyric slices for region r .

Symbol	Meaning and status
$F_s = f_\psi(W_s, H_s)$	Reproducible measurements from a declared analyzer, including loudness, spectra, estimated pitch, masking, and timing.
$I_t = (c_t, \xi_t, \tau_t, \rho_t)$	Intent: context, request text, listening horizon, and risk-exploration preference.
$Z_{u,k,t}$	Latent state of taste component k for listener u at time t .
$\alpha_{u,k,t}(I_t)$	Deterministic intent-conditioned activation weight computed from $Z_{u,k,t}$ and I_t .
$\Omega_{s,r} = (\pi_{s,r}, \eta_{s,r,1:L_\Omega})$	Persistent multimodal population response profile for region r ; π contains mode weights and η contains descriptor modes.
$\kappa_{u,s,r,t}$	Listener- and intent-conditioned response-mode assignment in $\{1, \dots, L_\Omega\}$.
$R_{u,s,r,t}$	Listener-conditioned perceptual and psychological response to region r .
$\bar{R}_{u,s,t}$	Aggregated song-level response used by observed behavior.
Γ_i	Latent semantic grounding of critique text T_i^{text} , including referenced region and response mode.
$E_{u,s,t}, Y_{u,s,t}^{\text{beh}}$	Exposure records whether and how song s was presented to listener u at t ; Y^{beh} records the response after exposure.
$\tilde{\phi}_{u,s,t}$	Augmented candidate feature vector containing musical and posterior-predictive response features.
$\mu_{u,k}, w_{u,k}, \Lambda_{u,k}, J_{u,k}$	Preferred center, monotone weights, diagonal range precision, and sparse zero-diagonal interaction matrix.
$\mathcal{O}$	Shared song, language, response, population parameters, and hierarchical hyperparameters.
$\mathcal{O}_u$	Listener-specific taste parameters.

The directed dependencies are summarized below. Each arrow is generative; dashed inference arrows are implemented by variational posteriors.

Generative and deterministic dependencies.

$$\begin{aligned}
X_s &\longrightarrow M_s, \\
X_s &\longrightarrow H_s, \\
(X_s, H_s) &\longrightarrow W_s, \\
X_s &\longrightarrow L_s, \\
(W_s, H_s) &\longrightarrow F_s \quad \text{deterministically,} \\
(X_{s,r}, H_s) &\longrightarrow \Omega_{s,r}, \\
(\Omega_{s,r}, X_{s,r}, Z_{u,1:K,t}, I_t) &\longrightarrow \kappa_{u,s,r,t}, \\
(\Omega_{s,r}, \kappa_{u,s,r,t}, X_{s,r}, W_{s,r}, L_{s,r}, Z_{u,1:K,t}, I_t) &\longrightarrow R_{u,s,r,t}, \\
\{R_{u,s,r,t} : r \in \mathcal{R}(H_s)\} &\longrightarrow \bar{R}_{u,s,t}, \\
(R_{u_i,s_i,r_i,t_i}, \Omega_{s_i,r_i}, X_{s_i,r_i}, Z_{u_i,1:K,t_i}, I_{t_i}) &\longrightarrow \Gamma_i, \\
(\Gamma_i, R_{u_i,s_i,r_i,t_i}, I_{t_i}) &\longrightarrow T_i^{\text{text}}, \\
(Z_{u,1:K,t}, I_t) &\longrightarrow \alpha_{u,1:K,t} \longrightarrow U_u, \\
(\text{policy}_t, \text{history}_t) &\longrightarrow E_{u,s,t}, \\
(E_{u,s,t}, \bar{R}_{u,s,t}, U_u, \text{session}_t) &\longrightarrow Y_{u,s,t}^{\text{beh}}.
\end{aligned}$$

Inference dependencies.

$$\begin{aligned} (W_s, L_s, M_s, F_s) &\dashrightarrow q_{\varphi_X}(X_s, H_s \mid W_s, L_s, M_s, F_s), \\ (\mathbb{T}, E, Y, I) &\dashrightarrow q_{\varphi}(\Omega, \kappa, R, Z, \Gamma \mid X, H, W, L, M, F, \mathbb{T}, E, Y, I). \end{aligned}$$

Definition 1 (Taste program). *For listener u ,*

$$T_u = (\mathcal{O}, \mathcal{O}_u, G_u, P_u). \quad (1)$$

The tuple contains shared parameters, listener-specific parameters, a dynamic factor graph, and a measurable randomized inference and sampling program.

Operational taste is

$$\text{Taste}_{u,t} = p(G_{u,t} \mid \mathcal{D}_{u,1:t}, I_t, \mathcal{O}, \mathcal{O}_u). \quad (2)$$

All random variables take values in standard Borel spaces, covering finite, countable, Euclidean, sequence, and finite-product domains.

3 Learned song representation

Let $\mathcal{R}(H_s)$ be the finite set containing one global region, all sections, and all passages induced by H_s . The model-relative representation is

$$\begin{aligned} X_s &= \left(X_s^{\text{global}}, \{X_{s,r}\}_{r \in \mathcal{R}(H_s)} \right), \\ X_{s,r} &= (X_{s,r}^{\text{acoustic}}, X_{s,r}^{\text{comp}}, X_{s,r}^{\text{struct}}, X_{s,r}^{\text{perform}}, \\ &\quad X_{s,r}^{\text{arrange}}, X_{s,r}^{\text{prod}}, X_{s,r}^{\text{lyric}}) \in \mathcal{X}. \end{aligned} \quad (3)$$

A generative model is

$$X_s \sim p_{\theta_X}(X), \quad H_s \sim p_{\theta_H}(H \mid X_s), \quad (4)$$

$$W_s \sim p_{\theta_W}(W \mid X_s, H_s), \quad L_s \sim p_{\theta_L}(L \mid X_s), \quad (5)$$

$$M_s \sim p_{\theta_M}(M \mid X_s), \quad F_s = f_{\psi}(W_s, H_s), \quad (6)$$

$$W_{s,r} = \text{crop}(W_s, r), \quad L_{s,r} = \text{align}(L_s, r). \quad (7)$$

The analysis encoder is

$$q_{\varphi_X}(X_s, H_s \mid W_s, L_s, M_s, F_s). \quad (8)$$

Proposition 1 (Latent reparameterization). *If $p_{\theta}(W) = \int p_{\theta}(W \mid X)p(X)d\nu(X)$ and g is a measurable bijection with measurable inverse, the pushforward prior under $X' = g(X)$ and decoder $p_{\theta'}(W \mid X') = p_{\theta}(W \mid g^{-1}(X'))$ produce the same waveform marginal.*

Proof. By the definition of pushforward measure,

$$\int p_{\theta'}(W \mid X')p'(dX') = \int p_{\theta}(W \mid g^{-1}(g(X)))p(dX) = p_{\theta}(W).$$

Commercial recordings therefore identify an equivalence class. Semantic coordinates require measurements, annotations, multiple views, restrictions, or interventions. $\square$

Multimodal region response profiles separate shared descriptions from listener-conditioned responses. Fix a finite upper bound L_Ω and let $\alpha_\Omega(X_{s,r}) \in \mathbb{R}_{>0}^{L_\Omega}$ be a learned positive concentration vector. Sparse concentrations let unsupported modes receive negligible mass:

$$\begin{aligned}\pi_{s,r} &\sim \text{Dirichlet}(\alpha_\Omega(X_{s,r})), & \eta_{s,r,\ell} &\sim p_{\theta_\eta}(\eta \mid X_{s,r}), \quad \ell = 1, \dots, L_\Omega, \\ \Omega_{s,r} &= (\pi_{s,r}, \eta_{s,r,1:L_\Omega}).\end{aligned}\tag{9}$$

For compact inference notation, $\Omega_s = \{\Omega_{s,r} : r \in \mathcal{R}(H_s)\}$, $\kappa_{u,s,t} = \{\kappa_{u,s,r,t} : r \in \mathcal{R}(H_s)\}$, and $R_{u,s,t} = \{R_{u,s,r,t} : r \in \mathcal{R}(H_s)\}$ denote the collections over all regions of song s . The latent assignment and region-level response are

$$\kappa_{u,s,r,t} \sim \text{Categorical}(\text{softmax}_\ell\{\log \pi_{s,r,\ell} + a_{\theta_\kappa}(\eta_{s,r,\ell}, X_{s,r}, Z_{u,1:K,t}, I_t)\}),\tag{10}$$

$$R_{u,s,r,t} \sim p_{\theta_R}(R \mid \eta_{s,r}, \kappa_{u,s,r,t}, X_{s,r}, W_{s,r}, L_{s,r}, Z_{u,1:K,t}, I_t).\tag{11}$$

For song-level behavior, aggregate region responses with intent-conditioned nonnegative weights:

$$v_{u,s,r,t} = \text{softmax}_r b_{\theta_v}(X_{s,r}, Z_{u,1:K,t}, I_t), \quad \bar{R}_{u,s,t} = \sum_{r \in \mathcal{R}(H_s)} v_{u,s,r,t} h_R(R_{u,s,r,t}).\tag{12}$$

For critique i associated with (u_i, s_i, r_i, t_i) ,

$$\Gamma_i \sim p_{\theta_\Gamma}(\Gamma \mid R_{u_i, s_i, r_i, t_i}, \Omega_{s_i, r_i}, X_{s_i, r_i}, Z_{u_i, 1:K, t_i}, I_{t_i}),\tag{13}$$

$$T_i^{\text{text}} \sim p_{\theta_T}(T \mid \Gamma_i, R_{u_i, s_i, r_i, t_i}, I_{t_i}).\tag{14}$$

The language inference network is

$$q_{\varphi_\Gamma}(\Gamma_i \mid T_i^{\text{text}}, X_{s_i, r_i}, W_{s_i, r_i}, L_{s_i, r_i}, \Omega_{s_i, r_i}, Z_{u_i, 1:K, t_i}, I_{t_i}).\tag{15}$$

The generative graph is acyclic: response profiles precede mode assignments and responses, grounding generates text, and the separate q distribution performs reverse inference. Critiques and exposure-corrected behavior update the posterior over Ω , κ , and R . Response modes are identifiable only up to permutation, so evaluation and posterior summaries must be permutation-invariant.

4 Intent and dynamic multi-taste state

Intent is

$$I_t = (c_t, \xi_t, \tau_t, \rho_t), \quad \rho_t \in \mathbb{R}.\tag{16}$$

Context c_t and request ξ_t influence features and taste activation; horizon τ_t determines sequence length; positive ρ_t is conservative and negative ρ_t encourages exploration.

Assumption 1 (Compact anchored augmented feature image). *The declared musical map $\phi^{\text{music}} : \mathcal{X} \times \mathcal{I} \rightarrow \Phi_{\text{music}}$ is continuous. The posterior-predictive augmented map*

$$\begin{aligned}\tilde{\phi}_{u,s,r,t} &= \phi^{\text{music}}(X_{s,r}, I_t) \oplus \mathbb{E}_q[h_R(R_{u,s,r,t})], \\ \tilde{\phi}_{u,s,t} &= \phi^{\text{music}}(X_s, I_t) \oplus \mathbb{E}_q[\bar{R}_{u,s,t}], \\ \phi^{\text{music}}(X, I) &= \phi_0(X) + \phi_{\text{ctx}}(X, I) + \phi_{\text{int}}(X, I).\end{aligned}\tag{17}$$

has image in a compact set $\tilde{\Phi} \subset \mathbb{R}^{\tilde{d}}$. Coordinates used in explanations are anchored by measurement protocols, supervision, calibrated listener reports, or declared model semantics.

For a hypothetical song state X , $\tilde{\phi}_u(X, I)$ uses the same construction with expectation under the generative posterior predictive $p_\theta(\Omega, R \mid X, Z_u, I)$ in place of the catalog posterior q .

Taste dynamics are hierarchical:

$$A_{u,k} \sim \mathcal{N}(A_k^{\text{pop}}, \Sigma_A), \quad Z_{u,k,t} = A_{u,k} Z_{u,k,t-1} + \epsilon_{u,k,t}, \quad (18)$$

$$\epsilon_{u,k,t} \sim \mathcal{N}(0, \Sigma_k^{\text{dyn}}), \quad \chi_{u,t} \sim \text{Bernoulli}(h_{u,t}). \quad (19)$$

A change point selects a larger innovation covariance. The active mixture is

$$\alpha_{u,k,t}(I_t) = \text{softmax}_k g_\omega(Z_{u,k,t}, c_t, \xi_t). \quad (20)$$

5 Preference energy and constraints

For every component, $\Lambda_{u,k}$ is diagonal positive semidefinite and $J_{u,k}$ is symmetric, sparse, and zero-diagonal. For catalog song s , define

$$U_u(X_s, I_t) = \sum_{k=1}^{K_u} \alpha_{u,k,t}(I_t) \left[w_{u,k}^\top \tilde{\phi}_{u,s,t} - \frac{1}{2} (\tilde{\phi}_{u,s,t} - \mu_{u,k})^\top \Lambda_{u,k} (\tilde{\phi}_{u,s,t} - \mu_{u,k}) + \frac{1}{2} \tilde{\phi}_{u,s,t}^\top J_{u,k} \tilde{\phi}_{u,s,t} - N_{u,k}(X_s, I_t) \right]. \quad (21)$$

The first quadratic form is a range penalty. The zero-diagonal quadratic form contains pairwise interactions only. The response coordinates let utility depend on several learned descriptions of a region while preserving listener- and intent-conditioned predictions. Soft negative regions use

$$N_{u,k}(X, I) = \sum_j \lambda_{u,k,j} \text{softplus}(g_{u,k,j}(X, I)). \quad (22)$$

Musical feasibility is

$$\mathcal{F}_u^{\text{music}}(I) = \{X : g_j^{\text{music}}(X, I) \leq 0\}. \quad (23)$$

Let $\mathcal{S}_u(I) \subseteq \mathcal{S}$ be the eligible indices in the already available catalog after applying artist exclusions and other metadata filters. This ranking-time set is external to the generative target.

Let $p_0(X \mid I)$ be a plausible-music density. The ideal-song density retains the original form after marginalizing the uncertain response profile and response through $\tilde{\phi}_u(X, I)$:

$$p_u^*(X \mid I, T_u) = \frac{p_0(X \mid I) e^{\beta_u(I) U_u(X, I)} \mathbf{1}[X \in \mathcal{F}_u^{\text{music}}(I)]}{Z_u(I)}, \quad (24)$$

$$Z_u(I) = \int_{\mathcal{F}_u^{\text{music}}(I)} p_0(X \mid I) e^{\beta_u(I) U_u(X, I)} d\nu(X). \quad (25)$$

Proposition 2 (Existence). *Under Assumption 1, finite parameters, continuous penalties, positive feasible base mass, and finite positive $\beta_u(I)$, the density in (24) exists.*

Proof. Continuity on compact $\tilde{\Phi}$ makes the posterior-predictive utility U_u bounded above by some finite $M(I)$. Hence $0 < Z_u(I) \leq e^{\beta_u(I) M(I)} < \infty$. Normalization gives unit integral. $\square$

A more general model can replace compactness by the direct integrability condition $Z_u(I) < \infty$.

6 Observations, exposure, and joint model

A parsed critique record contains

$$O_i = (u_i, s_i, r_i, t_i, j_i, p_i, \delta_i, \gamma_i), \quad (26)$$

where $r_i \in \mathcal{R}(H_{s_i})$ is the grounded region, j_i is the grounded feature or descriptor, p_i is polarity, δ_i is magnitude, and γ_i is parser confidence. The latent grounding Γ_i also carries posterior uncertainty over the region and response mode. Feature and pairwise observations are

$$Y_{u,s,r,j,t}^{\text{feat}} \sim \mathcal{N}(\omega_{u,j}^\top \tilde{\phi}_{u,s,r,t} - \zeta_{u,j}, \zeta_{u,j}^2), \quad (27)$$

$$Y_{u,s,s',t}^{\text{pair}} \sim \text{Bernoulli}(\text{logit}^{-1}(U_u(X_s, I_t) - U_u(X_{s'}, I_t))). \quad (28)$$

Exposure and behavior are

$$E_{u,s,t} \sim \text{Bernoulli}(e_\pi(s \mid \text{history}_t, \text{policy}_t)), \quad (29)$$

$$Y_{u,s,t}^{\text{beh}} \sim p_{\theta_Y}(Y \mid E_{u,s,t}, \bar{R}_{u,s,t}, U_u(X_s, I_t), \text{session}_t). \quad (30)$$

Including E prevents an unexposed song from being treated as negative evidence. Including $\bar{R}$ lets plays, skips, replays, saves, and completions update the response model directly.

Writing $\mathbb{T}$ for observed critique texts and suppressing deterministic Dirac factors, one acyclic joint factorization is

$$\begin{aligned} & p_\theta(X, H, W, L, M, \Omega, Z, \kappa, R, \Gamma, \mathbb{T}, E, Y \mid I) \\ &= p_\theta(X, H, W, L, M) p_\theta(\Omega \mid X, H) p_\theta(Z_0) \\ & \quad \times \prod_{t=1}^n p_\theta(Z_t \mid Z_{t-1}) \\ & \quad \times \prod_{u,s,r,t} p_\theta(\kappa_{u,s,r,t} \mid \Omega_{s,r}, X_{s,r}, Z_{u,t}, I_t) p_\theta(R_{u,s,r,t} \mid \kappa_{u,s,r,t}, \Omega_{s,r}, X_{s,r}, W_{s,r}, L_{s,r}, Z_{u,t}, I_t) \\ & \quad \times \prod_i p_\theta(\Gamma_i \mid R_{u_i, s_i, r_i, t_i}, \Omega_{s_i, r_i}, X_{s_i, r_i}, Z_{u_i, t_i}, I_{t_i}) p_\theta(T_i^{\text{text}} \mid \Gamma_i, R_{u_i, s_i, r_i, t_i}, I_{t_i}) \\ & \quad \times \prod_{u,s,r,j,t} p_\theta(Y_{u,s,r,j,t}^{\text{feat}} \mid X, \Omega, R, Z, I) \prod_{u,s,s',t} p_\theta(Y_{u,s,s',t}^{\text{pair}} \mid X, \Omega, R, Z, I) \\ & \quad \times \prod_{u,s,t} p_\theta(E_{u,s,t} \mid \text{history}_t, \text{policy}_t) p_\theta(Y_{u,s,t}^{\text{beh}} \mid E_{u,s,t}, \bar{R}_{u,s,t}, U_u, \text{session}_t). \end{aligned} \quad (31)$$

Proposition 3 (Program semantics). *If each node domain is standard Borel and each displayed conditional is a measurable Markov kernel in the stated topological order, the kernels define a unique joint measure on trajectory space.*

Proof. Order the kernels as (X, H, W, L, M) , deterministic measurements and slices, Ω, Z , deterministic $\alpha, \kappa, R, \Gamma, \mathbb{T}, E$, and Y . Every response-profile mixture has finite L_Ω , and every displayed conditional is a measurable Markov kernel by hypothesis. Repeated application of the Ionescu–Tulcea extension theorem gives a unique measure on trajectory space. Every finite factor graph is a marginal [4]. $\square$

7 Inference, learning, and retrieval

Define the observed data

$$\mathcal{D} = (W, L, M, F, \mathbb{T}, E, Y^{\text{feat}}, Y^{\text{pair}}, Y^{\text{beh}}). \quad (32)$$

A structured variational posterior

$$q_\varphi(X, H, \Omega, Z, \kappa, R, \Gamma \mid \mathcal{D}, I) \quad (33)$$

maximizes

$$\mathcal{L}_u(\theta, \varphi) = \mathbb{E}_{q_\varphi}[\log p_\theta(\mathcal{D}, X, H, \Omega, Z, \kappa, R, \Gamma \mid I)] - \mathbb{E}_{q_\varphi}[\log q_\varphi(X, H, \Omega, Z, \kappa, R, \Gamma \mid \mathcal{D}, I)]. \quad (34)$$

$$(\theta^*, \varphi^*) = \arg \max_{\theta, \varphi} \sum_u \mathcal{L}_u(\theta, \varphi). \quad (35)$$

The posterior over each $\Omega_{s,r}$ accumulates descriptions from critique groundings and exposure-corrected recommendation responses. Its several modes preserve population disagreement, while $\kappa_{u,s,r,t}$ and $R_{u,s,r,t}$ preserve listener and intent dependence.

For a transparent local prompt update, use a distinct pseudo-observation vector b_u :

$$w_u \sim \mathcal{N}(\mu_{u0}^w, \Sigma_{u0}^w), \quad (36)$$

$$b_u \mid w_u \sim \mathcal{N}(\mathbf{B}_u w_u, V_u). \quad (37)$$

The posterior is

$$\Sigma_u^w = ((\Sigma_{u0}^w)^{-1} + \mathbf{B}_u^\top V_u^{-1} \mathbf{B}_u)^{-1}, \quad (38)$$

$$\mu_u^w = \Sigma_u^w ((\Sigma_{u0}^w)^{-1} \mu_{u0}^w + \mathbf{B}_u^\top V_u^{-1} b_u). \quad (39)$$

These follow by completing the square.

Proposition 4 (Persistence of personalization). *For two listeners with common prior covariance and prompt observation model, define $K = \Sigma_0^w \mathbf{B}^\top (\mathbf{B} \Sigma_0^w \mathbf{B}^\top + V)^{-1}$. Then*

$$\mu_A^w - \mu_B^w = (I - K \mathbf{B})(\mu_{A0}^w - \mu_{B0}^w). \quad (40)$$

A difference outside the fully observed prompt subspace persists.

Proof. Use $\mu_u^w = \mu_{u0}^w + K(b - \mathbf{B} \mu_{u0}^w)$ and subtract the updates. $\square$

Fix utility scale per component by imposing $\|w_{u,k}\|_2^2 + (\text{tr}(\Lambda_{u,k})/\tilde{d})^2 + \|J_{u,k}\|_F^2 = 1$ and calibrating $\beta_u(I)$ separately. The toy setting $w = J = 0$ and $\Lambda = I_{\tilde{d}}$ satisfies this convention. The risk-sensitive catalog score is

$$S_u(s \mid I, \mathcal{D}_u) = \mathbb{E}_q[\beta_u(I) U_u(X_s, I)] - \rho(I) \text{SD}_q[\beta_u(I) U_u(X_s, I)] + \log \mathbf{1}[s \in \mathcal{S}_u(I)]. \quad (41)$$

The convention $\log 0 = -\infty$ enforces the external ranking filter. A playlist with $n(\tau) = \lceil \tau/\bar{d} \rceil$ items has

$$p(X_{1:n(\tau)} \mid I, T_u) \propto \prod_{m=1}^{n(\tau)} p_u^*(X_m \mid I, T_u) \prod_{m < n} \psi_{\text{div}}(X_m, X_n) \prod_{m=2}^{n(\tau)} \psi_{\text{seq}}(X_{m-1}, X_m, I). \quad (42)$$

8 Worked synthetic example

The prompt supplies positive evidence for breathiness and vocal-setting fit, negative evidence for cursive articulation, lingering, and masking, a preferred vocal-instrument balance, and an Ellie Goulding catalog exclusion.

Use a single-region, single-active-response-mode reduction of the augmented feature vector. This special case preserves the original numerical demonstration:

$$\tilde{\phi} = (B, G, N_C, N_L, V, P, E). \quad (43)$$

Let X_D denote *Diamonds* and X_L *After Love*. Synthetic posterior means are

$$\tilde{\phi}(X_D) = (.45, .80, .80, .55, .75, .95, .20), \quad \tilde{\phi}(X_L) = (.65, .85, .75, .75, .85, .25, .95). \quad (44)$$

Listener centers are

$$\mu_A = (.70, .90, .90, .80, .90, .90, .20), \quad \mu_B = (.70, .90, .90, .80, .90, .20, .95). \quad (45)$$

For this calculation set $\Lambda_A = \Lambda_B = I$, $w = 0$, $J = 0$, and common penalties to zero. Define the range energy, including the same half-factor as (21), by

$$d_u(X) = \frac{1}{2}(\tilde{\phi}(X) - \mu_u)^\top \Lambda_u (\tilde{\phi}(X) - \mu_u), \quad U_u(X, I) = -d_u(X). \quad (46)$$

The coordinate contributions and totals are shown in Table 2. With equal base densities and fixed-intent

Comparison	B	G	N_C	N_L	V	P	E	d_u
A, D	.03125	.00500	.00500	.03125	.01125	.00125	.00000	.08500
A, L	.00125	.00125	.01125	.00125	.00125	.21125	.28125	.50875
B, D	.03125	.00500	.00500	.03125	.01125	.28125	.28125	.64625
B, L	.00125	.00125	.01125	.00125	.00125	.00125	.00000	.01750

Table 2: Half-weighted squared-distance contributions.

$$\beta_A(I) = \beta_B(I) = 4,$$

$$\frac{p_A^*(X_D | I)}{p_A^*(X_L | I)} = e^{4(.50875 - .08500)} \approx 5.45, \quad (47)$$

$$\frac{p_B^*(X_L | I)}{p_B^*(X_D | I)} = e^{4(.64625 - .01750)} \approx 12.37. \quad (48)$$

The prompt constrains shared vocal dimensions while prior taste coordinates preserve cinematic-pop preference for A and melodic-electronic preference for B . All profiles are synthetic.

9 Faithful explanations

For coordinate j of the augmented feature vector, draw a replacement $\tilde{\phi}'_j$ from the conditional posterior predictive given retained coordinates, listener state, and intent. Define

$$\begin{aligned} \Delta_j(s, t) = & \mathbb{E}[U_u(\tilde{\phi}_{u,s,t}, I_t) | \mathcal{D}_u] \\ & - \mathbb{E}_{\tilde{\phi}'_j | \tilde{\phi}_{-j}, Z_u, I_t, \mathcal{D}_u}[U_u((\tilde{\phi}'_j, \tilde{\phi}_{-j}), I_t) | \mathcal{D}_u]. \end{aligned} \quad (49)$$

Exact conditional Shapley values sum to the utility difference between the candidate and declared conditional reference [5]. This follows because marginal contributions telescope for every ordering. Response-coordinate explanations integrate uncertainty over Ω , κ , and R . Explanations describe predictive, stated reasons unless controlled interventions support causal claims.

10 Equation guide

Every reference below resolves through a semantic label.

- (1), (2) Define the taste program and its current posterior state.
- (3) to (8) Separate global and regional song state, commercial observations, deterministic measurements, and analysis-time inference.
- (9) to (15) Define multimodal song-region response profiles, listener- and intent-conditioned mode assignment, regional perception, response aggregation, language grounding, and reverse inference.
- (16) to (20) Define operational intent fields, augmented posterior-predictive features, hierarchical dynamics, change points, and time-indexed taste activation.
- (21) to (25) Define range penalties, pairwise interactions, taste-specific negatives, musical feasibility and external catalog filters, and the normalized target density.
- (26) to (30) Define regional critique components, explicit feedback, exposure, response-aware behavior, and session context.
- (31) Gives the acyclic multiplicative joint factorization.
- (32) to (35) Define observed data, the augmented variational posterior, and its learning objective.
- (36) to (40) Give the local conjugate update and prove persistence of listener-specific priors.
- (41), (42) Use every intent field in uncertainty-sensitive retrieval and horizon-sensitive sequence generation.
- (43) to (48) Define and fully reproduce the corrected single-region, single-active-response-mode example.
- (49) Defines on-manifold counterfactual attribution for musical and posterior-predictive response features.

11 Identifiability, evaluation, and limitations

The song representation remains model-relative under Proposition 1. Semantic coordinates require anchors. Response-profile modes are identifiable only up to permutation, and weakly supported modes can merge, split, or receive negligible posterior mass. Utility scale is fixed by normalization, taste labels require permutation-invariant evaluation, and exposure propensities are recorded. Evaluation should include held-out likelihood, response-mode calibration, regional grounding, ranking, constraint violations, temporal adaptation, cold-start efficiency, counterfactual retrieval, and explanation faithfulness. Estimation of articulation, masking, perceptual response, and psychological response requires annotations, calibrated listener reports, prospective experiments, exposure correction, and abstention under weak posterior support. Feedback-driven response profiles also require audits for selection bias and population undercoverage.

12 Conclusion

The corrected program separates observed commercial recordings, global and regional song representations, multimodal population response profiles, listener- and intent-conditioned responses, language grounding, dynamic preferences, exposure, and retrieval. Critiques and exposure-corrected recommendation responses update several perceptual and psychological descriptions per song region. The target distribution remains proper under the augmented compactness conditions, and the worked example remains numerically consistent as a single-region, single-active-response-mode reduction.

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